

# CONCEPTUAL ELT DATA PIPELINE REPORT: BEEJAN TECHNOLOGIES

## 1. Executive Overview & Design Choice

Beejan Technologies receives customer complaints across four channels: social media, call center logs, SMS messages, and website forms. To unify these inputs, we designed an Extract, Load, Transform (ELT) data pipeline.

ELT is different from traditional methods because it stores raw data as soon as it comes in, before cleaning it up. This way, no data is lost, and it can handle big increases in traffic without a problem. Plus, it saves money by using cheap storage first and only paying for powerful computing when it's time to clean up the data. This approach is more efficient and cost-effective, making it a great option for handling large amounts of data. By storing raw data initially, ELT ensures that all information is preserved, and then it can be cleaned and processed as needed, reducing the risk of data loss and minimizing costs.

## 2. System Architecture & End-to-End Flow

- **Data Sources & Ingestion:** High-speed streams (social media, call center logs) are ingested continuously 24/7 through buffer queues to prevent system crashes. Lower-frequency files (SMS, web forms) are collected via automated hourly batch runs.
- **Hybrid Storage Architecture:**
  - o **Data Lake (Raw Reservoir):** Saves raw, uncleaned complaint data as-is immediately upon arrival for auditing and historical retention.
  - o **Data Warehouse (Structured Data):** Stores cleaned, structured tables ready for fast analytical querying and reporting.
- **Processing & Transformation:** Computations run inside the storage layer to clean duplicate entries, standardize formats, redact sensitive personal information (PII), tag complaint topics (Network, Billing, Service), and assign severity ratings (Low, Medium, High).
- **Serving & Analytics:** Cleaned tables power real-time Executive Dashboards (complaint trends and charts), trigger Automated Alerts when complaints spike, and support Ad-Hoc Queries for deep investigations.

## 3. Orchestration, Monitoring & DataOps Strategy

**Here's how it works:** there are these background tasks that run all the time, handling the constant stream of data. Then, every hour, some jobs kick in to load up big batches of data. Meanwhile, some automated tools are always keeping an eye on the pipeline, making sure everything is running smoothly. If something goes wrong, like a connection breaks, these tools send out instant notifications with logs to help figure out what's going on. This way, any issues can be spotted and fixed right away.

- **Cloud Hosting & Production Deployment:** The pipeline runs on flexible cloud infrastructure that automatically expands computing power during major outage surges. Updates to pipeline logic undergo automated testing in a safe environment before live release to ensure zero downtime.

## 4. Assumptions & Challenges

- **Assumptions:** All complaint channels allow automated data extraction, and incoming text uses standard languages.
- **Challenges & Mitigation:** Social media slang or sarcasm is managed by assigning confidence scores to tags, sending low-confidence items to human review. Traffic surges from major network outages are absorbed by ingestion buffers until processing catches up.